Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Investment Appraisal

Investment Appraisal

Investment appraisal is the analysis done to consider the profitability of an investment over the life of an asset alongside considerations of affordability and strategic fit.

Less resources and a greater number of proposals lead company owners and investors to decide which proposal to drop which to pick.

Profitability + Affordability + Strategic Fit Example

A software company is considering buying **new servers** for $5 million. Investment appraisal helps them calculate if the **increase in speed and efficiency** will earn enough profits to justify the cost, whether they can afford the upfront payment, and whether it aligns with their strategy (e.g., providing cloud hosting services).

Limited resources, many proposals – must choose wisely

- Proposal A: Launching a new mobile app for e-commerce.
- Proposal B: Expanding into AI-driven analytics software.
- Proposal C: Opening an office in Dubai to attract foreign clients.

Since resources are limited, appraisal is used to compare which project will give the highest return and fits best with the company’s future goals.

Life of an Asset Consideration

A logistics firm invests in **electric delivery vans**. Appraisal considers not only the **initial cost** but also savings on **fuel and maintenance** over 10 years, plus government tax incentives.

Investing

There is no single way of assessing and comparing the different proposals; factors that must be taken into consideration include, for example:

- the extent to which the proposals are consistent with the company's long-term plans;
- the risk attached to the proposals;
- the availability of the necessary resources even if the money is available.

DCF(Discounted cash flow)

Discounted cash flow (DCF) is a valuation method that **estimates the value of an investment** using its expected **future cash flows**. Analysts use **DCF to determine the value of an investment today**, based on projections of how much money that investment will generate in the future.

Discounted cash flow can help investors who are considering whether to acquire a company or buy securities. Discounted cash flow analysis can also assist business owners and managers in making capital budgeting or operating expenditures decisions.

DCF(Discounted cash flow)

It is important to realize that DCF is a tool that is used for many different purposes, for example:

- By investors on the stock market to assess whether the share price of a company reflects accurately its financial prospects;
- To assess whether it is better to purchase capital equipment or to lease it;
- To decide which of several possible projects is the most financially appealing;
- To decide whether a proposed capital project will be worthwhile.

THE TIME VALUE OF MONEY

- The time value of money is a basic financial concept that holds that money in the present is worth more than the same sum of money to be received in the future.

- This is true because money that you have right now can be invested and earn a return, thus creating a larger amount of money in the future.

- The time value of money is sometimes referred to as the net present value (NPV) of money.

Purchasing vs Leasing

Suppose you want to get a car worth Rs.10,00,000/- and you have that amount. The car finance department offers you to lease the car providing you pay a down payment of 20% that is Rs.2,00,000/- and a monthly payment of Rs.20,000 for 5 years(60 months).

Thus,

(20,000 * 60) + 2,00,000 = 14,00,000

While if you purchase it, you must pay Rs. 10,00,000/-

Would you purchase the car or lease it on these terms?

Discount factor

In general, if the interest rate is r, then the present value of a sum of money A due in t years time is:

$PV = A \times 1 \div (1 + r)^t$

The quantity $1 \div (1 + r)^t$ is known as the discount factor.

TABLE 8.1 *Discount factors for periods up to five years*

Interest Rate (%)	1	2	3	4	5
3	0.9709	0.9426	0.9151	0.8885	0.8626
4	0.9615	0.9246	0.8890	0.8548	0.8219
5	0.9524	0.9070	0.8638	0.8227	0.7835
6	0.9434	0.8900	0.8396	0.7921	0.7473
7	0.9346	0.8734	0.8163	0.7629	0.7130
8	0.9259	0.8573	0.7938	0.7350	0.6806
9	0.9174	0.8417	0.7722	0.7084	0.6499
10	0.9091	0.8264	0.7513	0.6830	0.6209
15	0.8696	0.7561	0.6575	0.5718	0.4972
20	0.8333	0.6944	0.5787	0.4823	0.4019

Calculating values from DCF table

Discount factor for a discount rate of 8 percent over a period of four years is 0.7350. This means that, if the discount rate is 8 per cent, the present value of a sum of £1,000 payable in four years time is **£1000 × 0.7350 = £735**

Example

A new van will cost £10,000. There will be annual costs of £500 for insurance and £150 for road tax. The cost of maintenance is estimated to be £200 in each of the first two years, £300 in year 3, £400 in year 4 and £500 in year 5.

At the end of the fifth year, it is expected that the van will be sold for around £2,000. The interest rate that the company pays on its borrowings is 10 percent.

Example (Cont.)

If the van is hired(rented), Van hire costs £35 per day and it hires a van for about 100 days a year. All the costs are subject to inflation, which is judged to be around 5 per cent over the period, but the resale value of the van is the cash figure expected at the time. Which one would be better approach buying a van or renting?

Example (Solution)

Item	Detail
Van cost	£10,000 upfront
Insurance	£500 per year
Tax	£150 per year
Maintenance	£200 (Y1–Y2), £300 (Y3), £400 (Y4), £500 (Y5)
Resale value	£2,000 at end of Year 5
Inflation	5% per year
Discount rate (interest)	10%
Van hire cost	£35/day × 100 days = £3,500 per year (subject to 5% inflation)

Step 1: Adjust for inflation

All **costs except the resale value** increase by 5% per year. So, we calculate the inflated annual costs:

Year	Insurance + Tax	Maintenance	Hire cost
1	$650 \times 1.05 =$ **£683**	$£200 \times 1.05 =$ **£210**	$£3{,}500 \times 1.05 =$ **£3,675**
2	$650 \times 1.05^2 =$ **£717**	$200 \times 1.05^2 =$ **£331**	$3{,}500 \times 1.05^2 =$ **£3,859**
3	$650 \times 1.05^3 =$ **£752**	$300 \times 1.05^3 =$ **£463**	$3{,}500 \times 1.05^3 =$ **£4,052**
4	$650 \times 1.05^4 =$ **£790**	$400 \times 1.05^4 =$ **£608**	$3{,}500 \times 1.05^4 =$ **£4,254**

Step 2: Compute annual **cash flows** for buying

Each year's cash flow = negative of (Tax + Insurance + Maintenance), except the final year includes **sale proceeds**

Year	Components	Annual Cash Flow
0	Purchase (£10,000) + Year 0 insurance/tax/maintenance	**–10,850**
1	–(683 + 210)	**–893**
2	–(717 + 331)	**–1,048**
3	–(752 + 463)	**–1,215**
4	+Sale £2,000 – (790 + 608)	**+602**

Step 3: Discount each year's cash flow (NPV of annual flow)

Add up all the Present Values

Year	Cash Flow	DF	PV (Cash Flow × DF)
0	–10,850	1.000	–10,850
1	–893	0.909	–812
2	–1,048	0.826	–866
3	–1,215	0.751	–914
4	+602	0.683	+412

Add up all the Present Values:

Total NPV: (-10,850) + (-812) + (-866) + (-914) + 412 = –13,030

Step 4: Continuing to rent

Year	Rent cost	PV factor (10%)	NPV
0	–3,500	1.000	–3,500
1	–3,675	0.909	–3,341
2	–3,859	0.826	–3,189
3	–4,052	0.751	–3,044
4	–4,254	0.683	–2,906

Add up all the Present Values:

Total NPV: (-3,500) + (-3,341) + (-3,189) + (-3,044) + (-2,906) = –15,980

Step 5: Compare NPVs

Option	**Total NPV**	**Better choice**
Buy van	–£13,030	Lower cost
Rent van	–£15,980	More expensive

TABLE 8.2 *DCF analysis of van purchase versus leasing*

	Year 0	Year 1	Year 2	Year 3	Year 4
Buying a van					
Van purchase/sale	(10000)				2000
Tax and insurance	(650)	(683)	(717)	(752)	(790)
Maintenance	(200)	(210)	(331)	(463)	(608)
Annual cash flow	(10850)	(893)	(1048)	(1215)	602
NPV of annual flow	(10850)	(812)	(866)	(914)	412
Total NPV	(13030)				
Continuing to rent					
Annual costs	(3500)	(3675)	(3859)	(4052)	(4254)
NPV of annual costs	(3500)	(3341)	(3189)	(3044)	(2906)
Total NPV	(15980)				

Cost of capital

- Even if the company has the cash available to buy the van outright, there is still a cost because the company will lose the income it could have received by investing the money somewhere else, in a suitable interest-bearing account. Such a cost is known as an opportunity cost.

- Opportunity cost is the cost of the investment/opportunity you miss out on over the cost of the investment /opportunity you choose.

- If the company can pay cash for the van, this is the interest rate it would be appropriate to use in the DCF analysis.

Inflation

- Inflation in a financial context means the fall in the value of money over time.

- It is usually expressed as an annual percentage.
- Thus, for example, an inflation rate of 5 per cent means that in a year’s time goods that today cost £100 will cost £105. In two years, they will cost £100 × 1.05 × 1.05 = £110.25.

Handling Inflation

- Initially estimate all costs in today’s money.
- Then estimate an inflation rate of 5 per cent and adjusted the cash flows for future years to take this into account.
- Used the ‘monetary’ rate of interest rather than the ‘real’ rate.
- In normal economic conditions this is the simplest way to carry out a DCF analysis.